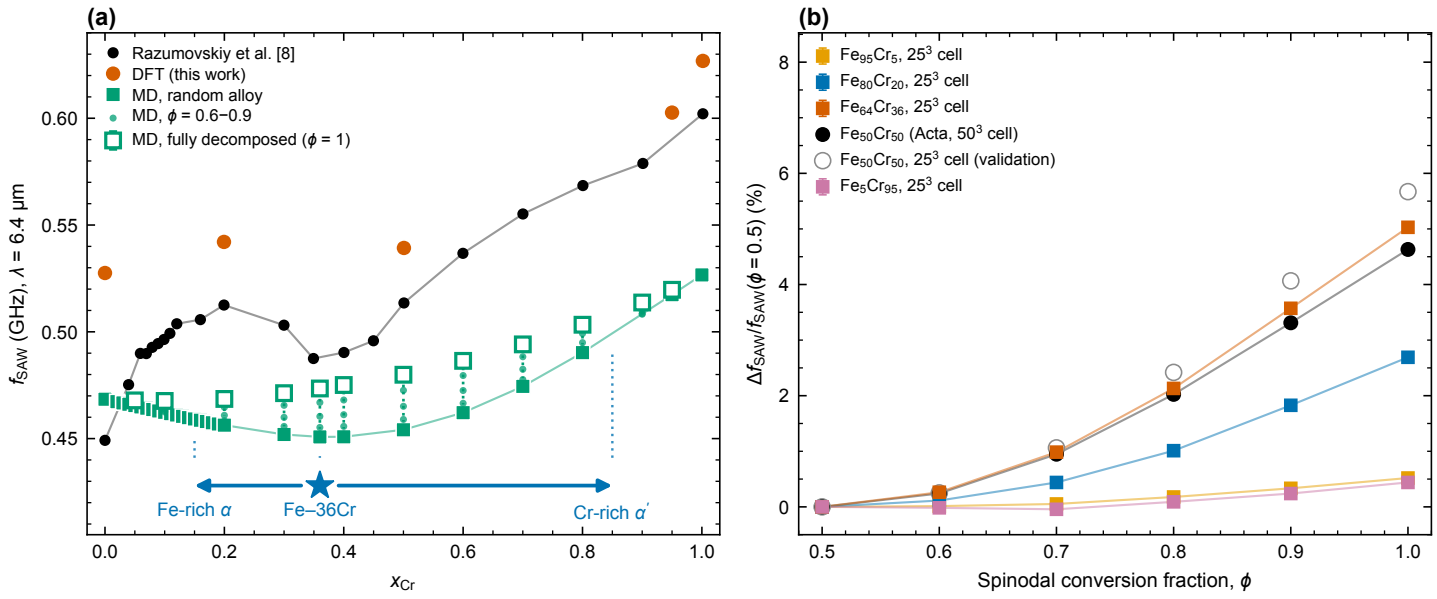

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, ϕ = 0.6–0.9
MD, fully decomposed (ϕ = 1)
f_SAW (GHz), λ = 6.4 μm
0.45
0.50
0.55
0.60
0.0
0.2
0.4
0.6
0.8
1.0
x_Cr
Fe-rich α
Fe–36Cr
Cr-rich α′
(b)
Fe95Cr5, 25^3 cell
Fe80Cr20, 25^3 cell
Fe64Cr36, 25^3 cell
Fe50Cr50 (Acta, 50^3 cell)
Fe50Cr50, 25^3 cell (validation)
Fe5Cr95, 25^3 cell
Δf_SAW/f_SAW(ϕ = 0.5) (%)
0
2
4
6
8
0.5
0.6
0.7
0.8
0.9
1.0
Spinodal conversion fraction, ϕ